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RESEARCH ARTICLE



Analysis of Sentinel-1 and Sentinel-2 data for post-earthquake reconstruction monitoring: a case study of the 2024 Noto Peninsula earthquake

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ABSTRACT

Sentinel-1 and Sentinel-2 have been widely used for earthquake damage detection; however, their use in monitoring long-term recovery activities remains largely unexplored. This study evaluates the effectiveness of Sentinel-1 and Sentinel-2 satellite observations for monitoring post-earthquake recovery processes. Focusing on the 2024 Noto Peninsula earthquake, indices were used to assess the detectability of surface changes associated with two key recovery activities: debris removal and temporary housing construction. A detailed analysis of the long-term Sentinel-1 backscatter time-series monitoring was conducted. Public demolition sites exhibited characteristic temporal changes in the vertical transmit–horizontal receive polarization: an initial increase associated with debris generation, followed by a gradual decrease, and then a sharp decline corresponding to the commencement of demolition. At the temporary housing construction sites, the vertical transmit–vertical receive polarization values increased rapidly during construction and remained elevated. The findings revealed that different recovery activities produce distinct backscatter signatures, suggesting that synthetic aperture radar time-series analysis can serve as an effective tool for characterizing post-disaster recovery. Applying these temporal patterns across wider areas enables the identification of collapsed structures and newly constructed temporary housing. This study demonstrated the potential of freely available satellite data to systematically monitor reconstruction processes.

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Sentinel-1; Sentinel-2; post-earthquake recovery monitoring; backscatter coefficients; Noto Peninsula earthquake; time-series analysis

Introduction

In 2024, 393 disasters were recorded in the Emergency Events Database (EM-DAT), a global disaster database maintained by the Centre for Research on the Epidemiology of Disasters (CRED). These disasters resulted in 16,753 deaths and economic losses of USD 241.95 billion (CRED, 2025). In response to such risk, the Sendai Framework for Disaster Risk Reduction emphasizes not only immediate disaster response but also long-term disaster risk management, a central principle of which is “Build Back Better” (BBB), which promotes the strengthening of disaster resilience through recovery processes (Mizutori, 2020; UNISDR, 2015).

To effectively incorporate lessons learned from past recovery efforts in future reconstruction planning, it is essential to assess the extent to which BBB principles are achieved. However, such assessments require objective and quantifiable indicators of recovery, which are currently insufficiently developed (Fernandez & Ahmed, 2019; Zhou et al., 2022). While a comprehensive BBB assessment encompasses physical, social, economic, and psychological dimensions (Fernandez & Ahmed, 2019; Kim & Olshansky, 2014), this study focused specifically on observable physical changes in buildings as a measurable component of the recovery process.

Satellite remote sensing offers a means of monitoring the recovery of a built environment over wide areas at regular intervals, thereby supplementing traditional ground surveys (Platt et al., 2016). Numerous

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studies have used pre- and post-event satellite data to detect and assess earthquake damage (Yu et al., 2024; Zhu et al., 2024). High-resolution imagery enables detailed damage detection (Adriano et al., 2021; Cho et al., 2023; Nonaka et al., 2021; Qing et al., 2022; Wang et al., 2022; Zheng et al., 2021), but its temporal availability is limited and commercial data are often costly. Therefore, freely accessible and regularly acquired datasets, such as the European Space Agency's Sentinel series, offer substantial advantages despite their reduced spatial resolution. The Sentinel series, which comprises the Sentinel-1 synthetic aperture radar (SAR) data and Sentinel-2 optical data, provides continuous and complementary observations (Drusch et al., 2012; Torres et al., 2012). Recent studies have demonstrated the use of Sentinel-2 indices, such as the Normalized Difference Built-up Index (NDBI) and Sentinel-1 amplitude and coherence, for earthquake damage mapping in Turkey and Syria (Yu et al., 2024). Multi-temporal coherence has been used for building damage detection (Liu et al., 2024), while amplitude differences and phase coherence have been applied to urban damage assessment (Wang et al., 2023). Similarly, (Serifoglu Yilmaz et al., 2023) combined Sentinel-2 indices and Sentinel-1 backscatter data to evaluate the impact of earthquakes.

However, although both Sentinel-1 and Sentinel-2 have been proven to be effective for earthquake damage mapping, their application in monitoring long-term post-disaster recovery remains limited. (Lamboglia et al., 2025) employed Sentinel-2-derived luminance and a Normalized Difference Vegetation Index (NDVI) to track recovery dynamics, demonstrating the potential of optical data for recovery assessment. However, Sentinel-1 data have rarely been explored for long-term recovery monitoring, and the temporal behavior of SAR-derived indicators associated with reconstruction activities has not been examined. Therefore, further investigations are required to evaluate the utility of these freely available satellite datasets for tracking post-disaster recovery processes.

This study addresses this gap by analyzing both Sentinel-2 optical indices and Sentinel-1 SAR data to monitor post-earthquake recovery. Focusing on the 2024 Noto Peninsula earthquake, we examined the detectability of surface changes associated with two key recovery activities: debris removal under a public demolition program and the construction of temporary housing. We first compared multiple indicators derived from both satellites at sites where recovery activities occurred against nearby reference areas with similar pre-earthquake land-use conditions. We then selected the most effective indicators and assessed their applicability across a wider area. This study contributes to the development of satellite-based methods to track observable physical recovery processes in a scalable and cost-effective manner.

Materials and methods

The 2024 Noto peninsula earthquake

At 16:10 JST on January 1, 2024, a magnitude 7.5 earthquake struck the Noto Peninsula, in the northern part of Ishikawa Prefecture in central Japan (USGS, 2024). The earthquake triggered multiple cascading hazards, including liquefaction, tsunamis, landslides, crustal uplift, and fires. These impacts were exacerbated by severe winter conditions, resulting in widespread damage (Suppasri et al., 2024). A total of 551 fatalities and damage to 6,273 buildings have been reported (CRED, 2025; IPRED, 2024). Wajima and Suzu cities experienced particularly severe impacts; therefore, these areas were selected as the focus of this study. To capture both the pre-event baseline and post-event recovery processes, the observation period was set from July 2023, six months before the earthquake, to June 2025, one and a half years after the earthquake.

Target sites

Two sites in the earthquake-affected area were selected, each representing a distinct phase of the post-disaster response: publicly funded demolition in Wajima City and temporary housing construction in Suzu City. To characterize the changes captured in the satellite data at each target site, a corresponding control site was selected for comparison based on the following criteria: (1) spatial proximity to the target site, ensuring similar environmental and meteorological conditions; (2) similar land cover or land use characteristics before the event of interest; and (3) no direct impact from the event being monitored. This paired-comparison approach enabled the changes observed at the target site to be distinguished from seasonal or other background variations captured at the control site.

Publicly funded demolition in Wajima city

In Wajima City, the damage was particularly severe in the area known as the Morning Market Street, a densely built district of wooden houses (IPRED, 2024; Nishino, 2025). A fire, believed to have been caused by an earthquake-triggered electrical failure, destroyed approximately 49,000 m² of land and resulted in the loss of 240 buildings (Fire & Disaster Management Investigation Commission Based on the Great Wajima City Fire, 2024). Subsequently, publicly funded demolition was carried out from June 5, 2024, to April 25, 2025 (NHK, 2025; The Hokkoku Shimbun, 2024). Although rubble and partially burned structures initially remained, debris was gradually removed until the affected area was cleared.

Based on these criteria, the fire-destroyed district was defined as the target site, and a comparable urban area located across the river was designated as the control site (Figure 1, Table 1).

Construction of temporary housing in Suzu city

Suzu City was the municipality closest to the earthquake's epicenter and consequently suffered severe damage to buildings and roads. A tsunami with a height of approximately 5 m reached the city within one



Figure 1. Target sites in Wajima City, with the target site shown in red and the control site in blue. Image © 2023 Airbus, Google Earth.

Table 1. Coordinates of target and control sites.

Location	Longitude (°)	Latitude (°)
Wajima (Target)	136.90118	37.39711
Wajima (Control)	136.89907	37.39804
Suzu (Target)	137.24158	37.39703
Suzu (Control)	137.24254	37.39757

minute after the earthquake (IPRED, 2024; Suppasri et al., 2024). Due to extensive housing damage, temporary housing was constructed in Suzu for the affected residents. Among these projects, the temporary housing designed by Shigeru Ban was particularly notable (IPRED, 2024). The structures are two-story wooden buildings constructed using dowel-laminated timber (DLT), a type of engineered timber material that does not require large-scale manufacturing facilities (Nikkei Business Publications, 2024). Construction began on March 1, 2024, approximately two months after the earthquake, with two units completed by the end of June 2024, and two additional units completed in July 2024 (Nikkei Business Publications, 2024; Yoshida, 2024).

The temporary housing construction site was defined as the target site, whereas the adjacent area across the road, which had a similar grassland before construction, was designated as the control site (Figure 2, Table 1).

Sentinel-2 data

Sentinel-2 is an optical Earth observation mission operated by the European Space Agency (ESA) that provides multispectral imagery at a spatial resolution of 10–60 m and a revisit frequency of five days at the

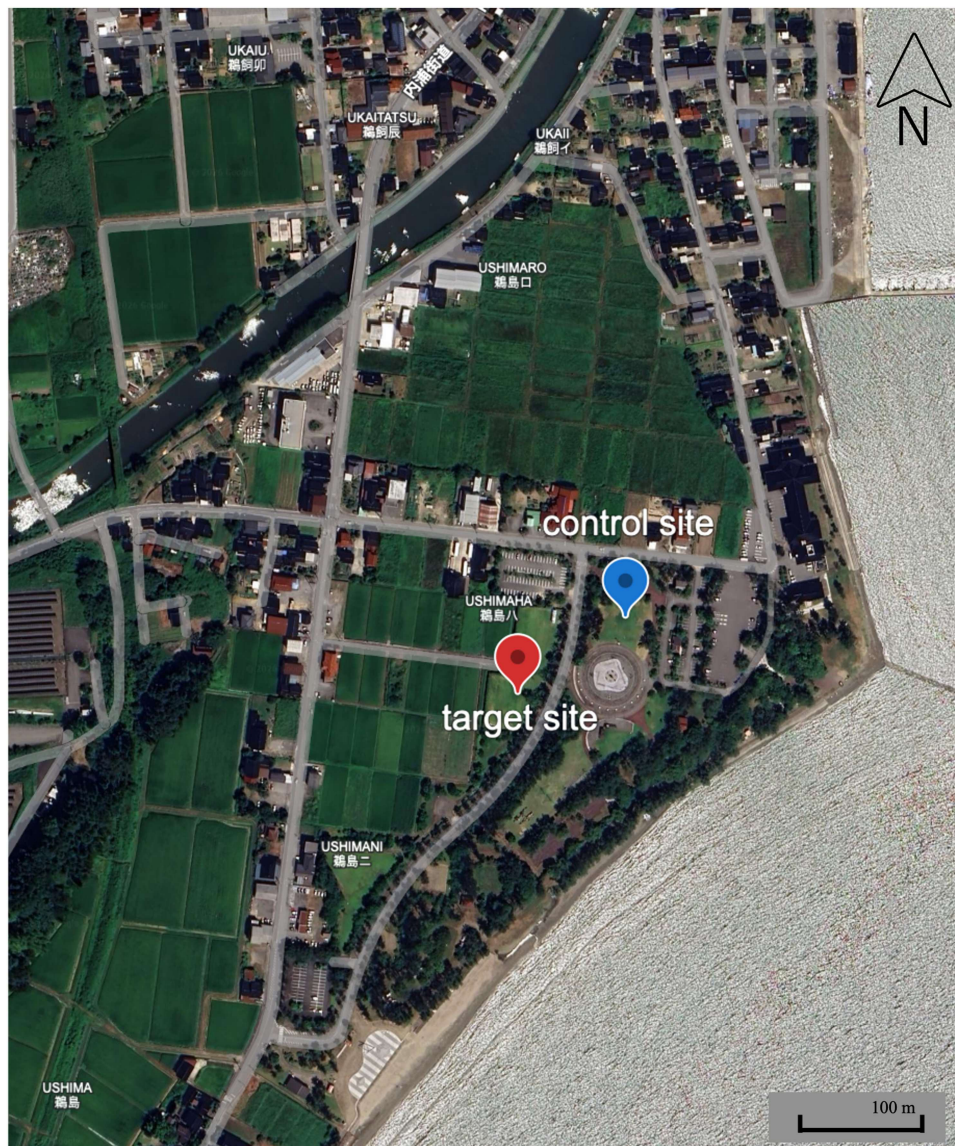


Figure 2. Target sites in Suzu City, with the target site shown in red and the control site in blue. Image © 2023 Airbus, Google Earth.

equator (Drusch et al., 2012). For this study, Level-2A bottom-of-atmosphere reflectance products (S2MSI2A) covering the study area within tile ID T53SPB were obtained (CDSE, n.d.).

Three spectral indices were employed to evaluate post-earthquake changes in urban areas. Perceived Lightness (PL) and NDVI were selected based on the work of (Lamboglia et al., 2025), who demonstrated the effectiveness of luminance and vegetation indices in tracking post-disaster recovery dynamics using Sentinel-2 data. NDBI was also included due to its widespread application in earthquake damage assessment studies (Serifoglu Yilmaz et al., 2023; Yu et al., 2024) and its ability to capture changes in the built environment associated with demolition and construction activities.

However, because Sentinel-2 is an optical satellite, its imagery is highly affected by clouds and atmospheric conditions, and many acquired scenes cannot be directly used for analysis. To address this, a Scene Classification Map (SCL) provided by the ESA was used to exclude observations affected by snow, clouds, cloud shadows, or missing data (Sentinel Hub, n.d.). When either the target or the control site fell into one of these categories, the corresponding observations were excluded from the analysis. During the study period, 331 scenes were downloaded. However, more than 80% were unusable for analysis because of missing data or cloud contamination. Even among scenes without missing data, approximately 70% were affected by clouds or cloud shadows.

In addition, visual inspection of the RGB composites revealed scenes with clearly anomalous values compared with other images from the same period. To exclude such anomalies objectively, we employed an outlier detection method based on the rolling Median Absolute Deviation (MAD). To remove outliers that could affect the analysis results in the area surrounding the target sites, a $1 \text{ km} \times 1 \text{ km}$ area centered on each target site was defined as the analysis unit, and the indices (NDVI, NDBI, and PL) were extracted. For each image, the mean and standard deviation of valid pixels were calculated and used as representative values. Outlier detection was then performed on the time series of the representative values. Specifically, a moving window of seven observations was applied, within which the local median was used as a reference and the MAD was calculated. Each observation was then standardized using the local MAD to obtain a robust z-score, and observations with a robust z-score greater than five were removed as outliers. Consequently, approximately 20 scenes were excluded from each site, allowing us to remove the influences of clouds and snow that were not filtered out by the SCL. Ultimately, approximately 30 scenes were analyzed.

Perceived lightness

PL is a parameter that quantifies changes in pixel brightness derived from the standard RGB (sRGB) color space (Zhu et al., 2015) and has recently been applied to urban change detection in post-disaster contexts (Lamboglia et al., 2025). This procedure was performed as described by (Lamboglia et al., 2025).

First, Sentinel-2 red (Band 4), green (Band 3), and blue (Band 2) values, each provided at a 10 m spatial resolution (CDSE, n.d.), were converted to sRGB as follows:

$$V = \begin{cases} \frac{V_0}{12.92} & V_0 \leq 0.04045, \\ \left(\frac{V_0 + 0.055}{1.055} \right)^{2.4} & V_0 > 0.04045, \end{cases} \quad (1)$$

where V_0 is the red, green, or blue value and V is the standard red, green, or blue value (Lamboglia et al., 2025), following the IEC 61966-2-1 standard (International Electrotechnical Commission, 1999).

Luminance Y is then calculated as

$$Y = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B, \quad (2)$$

where R , G , and B are the standard red, green, and blue values obtained from Equation (1), respectively.

Following the standard formulation (Zhu et al., 2015), PL is then defined as

$$PL(Y) = \begin{cases} 116 \cdot Y^{1/3} - 16 & Y > 0.01, \\ 903.3 \cdot Y & Y \leq 0.01, \end{cases} \quad (3)$$

NDVI

NDVI is a widely used indicator of vegetation activity and recovery, defined as

$$NDVI = \frac{NIR - Red}{NIR + Red}, \quad (4)$$

where near-infrared (NIR) and red denote the reflectance of the near-infrared and red bands, respectively (Lamboglia et al., 2025). In the Sentinel-2 imagery, these correspond to bands 8 (NIR, 10 m resolution) and 4 (red, 10 m resolution) (CDSE, n.d.).

NDBI

NDBI has been used to monitor built-up structures and impervious surfaces (Yu et al., 2024; Zha et al., 2003). In Sentinel-2, these correspond to the short-wavelength infrared (SWIR) (band 11, 20 m resolution) and NIR (band 8, 10 m resolution) (CDSE, n.d.). It is defined as

$$NDBI = \frac{SWIR - NIR}{SWIR + NIR}, \quad (5)$$

Because the spatial resolutions differed, the SWIR band was resampled to 10 m to enable consistent pixel-wise analysis.

Sentinel-1 data

Sentinel-1 is a SAR mission that provides C-band observations in vertical transmit–vertical receive (VV) and vertical transmit–horizontal receive (VH) polarizations, with a native spatial resolution of 10 m, a 12-day repeat cycle, and the capability to acquire observations independent of weather conditions and time of day (Torres et al., 2012). In this study, VV and VH backscatter amplitudes were employed as indicators because changes in backscatter have been shown to reflect structural damage (Serifoglu Yilmaz et al., 2023; Wang et al., 2023). Coherence was also included because it is sensitive to surface changes (Liu et al., 2024; Wang et al., 2023). By applying these indicators over an extended observation period, this study explored their potential for tracking post-disaster recovery processes.

Data covering the Noto Peninsula were selected from relative orbit number 17, specifically focusing on the frames corresponding to ID T2059. Data were available at 12-day intervals throughout the study period. The processed products were obtained via a Hybrid Pluggable Processing Pipeline (HyP3), a cloud-based SAR processing service provided by the Alaska Satellite Facility (ASF) (ASF, n.d.; Hogenson et al., 2020). These included radiometrically terrain-corrected backscatter coefficients at 10 m resolution and coherence maps generated with a 10×2 multi-look configuration, which reduced speckle noise at the cost of spatial detail, yielding an effective resolution on the order of several tens of meters (ASF, n.d.; ESA, n.d.; Hogenson et al., 2020).

VV and VH backscatter amplitude

The differences between the VV and VH polarizations primarily reflect the interaction of the radar signals with surface structures. VV polarization is particularly sensitive to double-bounce scattering effects, typically produced by buildings and other vertical features, whereas VH polarization responds more strongly to volume scattering, such as that produced by vegetation canopies (Delgado Blasco et al., 2020; Patruno et al., 2020). Using both the VV and VH channels captures different scattering characteristics, enabling the detection of structural changes induced by earthquakes or subsequent reconstruction.

To improve the robustness of the analysis, backscatter coefficients were computed using a 3×3 local window, which helps to reduce speckle noise while retaining essential structural information. This approach allowed for a more stable time-series interpretation of the surface changes in the affected areas.

Coherence

Interferometric coherence quantifies the similarity between two SAR acquisitions and is expressed as the normalized cross-correlation of their complex signals (Jacob et al., 2020). It can be written as:

$$\gamma = \frac{|E[s_1 s_2^*]|}{\sqrt{E[|s_1|^2] E[|s_2|^2]}} \quad (6)$$

where s_1 and s_2 denote the complex SAR observations at the two acquisition times and E represents the expectation operator (López-Martínez & Pottier, 2007; Touzi et al., 1999).

A reduction in coherence is typically associated with surface disturbances, vegetation dynamics, or structural damage. In this study, the coherence was calculated for each acquisition using the temporally closest preceding and succeeding images on the HyP3 platform (ASF, n.d.), thereby enabling a time-series analysis of surface changes.

Results

Indicator comparison for demolition in Wajima city

Sentinel-2 indicators for demolition

Figure 3 shows the three Sentinel-2 indices at the target and control sites in Wajima. Table 2 summarizes the mean difference and standard deviation (SD) ratio between the two sites for each indicator across the pre-earthquake (Pre-EQ), post-earthquake to demolition onset (Post-EQ), and public demolition (Demolition) periods. Values for the post-demolition period (Post-Demolition) are also included for reference. However, as only two observations are available, they are not discussed further in this section.

The most pronounced difference was observed in NDBI values. Before the earthquake, both sites had nearly identical values. Although the two pre-event observations (November and December 2023) showed some discrepancies, the mean difference between the target and control sites was only 0.008. After the earthquake, NDBI values at the target site decreased markedly, and the mean difference exceeded 0.1 throughout the post-earthquake and demolition periods (− 0.122 and − 0.113, respectively). This result indicates the disappearance of artificial structures (i.e. buildings). However, we did not observe a sustained downward trend during the demolition period. The intersite difference narrowed at several points, notably around late April 2024 and again from April 2025 onward, likely reflecting the influence of seasonal variability.

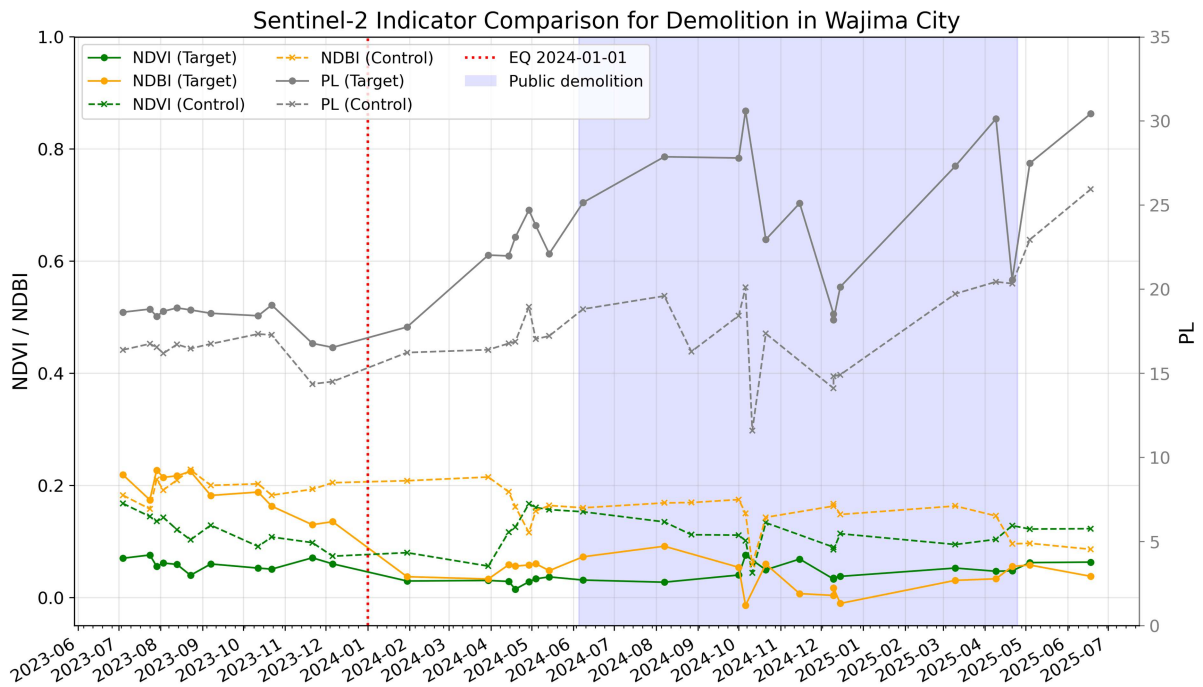


Figure 3. Sentinel-2 indices (NDBI, PL, NDVI) at the target and control sites in Wajima.

Table 2. Comparison of Sentinel-2 indicators between target and control sites for the demolition site in Wajima City. $\bar{x}_T - \bar{x}_C$ denotes the difference between the target and control means. SD_T/SD_C denotes the ratio of their standard deviations.

Indicator	Period	$\bar{x}_T - \bar{x}_C$	SD_T/SD_C
NDVI	Pre-EQ	−0.060	0.37
NDVI	Post-EQ	−0.094	0.16
NDVI	Demolition	−0.063	0.54
NDVI	Post-Demolition	−0.060	1.25
NDBI	Pre-EQ	−0.008	1.92
NDBI	Post-EQ	−0.122	0.32
NDBI	Demolition	−0.113	1.02
NDBI	Post-Demolition	−0.043	1.89
PL	Pre-EQ	2.023	0.86
PL	Post-EQ	5.137	2.46
PL	Demolition	7.104	1.54
PL	Post-Demolition	4.516	0.98

The PL index also exhibited clear intersite differences. Although the target site exhibited higher values than the control site even prior to the earthquake, the difference widened thereafter; the mean difference increased from 2.023 in the pre-earthquake period to 5.137 in the post-earthquake period and to 7.104 during the demolition period. By late April 2025, the two sites converged to nearly identical values; however, at most other times, the target site remained brighter, a pattern consistent with the transition from built structures to bare ground. Pronounced declines in the target site were observed in October and December 2024, whereas the control site exhibited lower values during these periods. This suggests possible seasonal and observational effects. However, no usable images were available for November 2024 or January and February 2025, making it difficult to draw firm conclusions.

For NDVI, the control site consistently maintained higher values throughout the study period, whereas the target site exhibited little variation. This contrast was reflected in the SD ratio (SD_T/SD_C), which ranged from 0.16 to 0.54 across the pre-earthquake to demolition periods, indicating that the temporal variability at the target site was substantially smaller than that at the control site. Even before the earthquake, higher values were evident at the control site during summer months, likely reflecting the presence of vegetation. The absence of pronounced fluctuations at the target site is consistent with the interpretation that the residential area has transitioned to bare ground. However, NDVI is unable to capture the progression of the demolition process.

Sentinel-1 indicators for demolition

Figure 4 shows the VV and VH backscatter coefficients for the target and control sites in Wajima. Table 3 summarizes the period-wise comparison of Sentinel-1 indicators between the two sites, following the same format as that for Sentinel-2.

The most notable feature is the change in the VH polarization. Prior to the earthquake, the target site exhibited slightly higher values than the control site, with a mean difference of 1.29 dB. Following the earthquake, the VH values at the target site increased markedly, and the mean intersite difference rose to 4.53 dB. Before the onset of the publicly funded demolition period, these values showed a gradual decline. However, the standard deviation ratio increased from 1.32 to 1.58 relative to the pre-earthquake period, indicating greater variability at the target site. During the demolition period, VH values decreased markedly at the target site, in contrast to the relatively stable values at the control site; the mean difference reversed to − 5.22 dB and the standard deviation ratio reached 2.51, further reflecting the magnitude of change at the target site. Although the post-demolition period comprised only five observation dates, the target site mean remained 4.67 dB below that of the control site. Considering that VH polarization is sensitive to volume scattering, this pattern is consistent with earthquake-induced fires generating debris that initially enhanced VH backscatter, followed by a decline as the debris and remaining structures were progressively removed during demolition.

The VV polarization exhibited a similar trend, capturing the progression of the publicly funded demolition process. However, compared with VH, the changes at the target site relative to the control

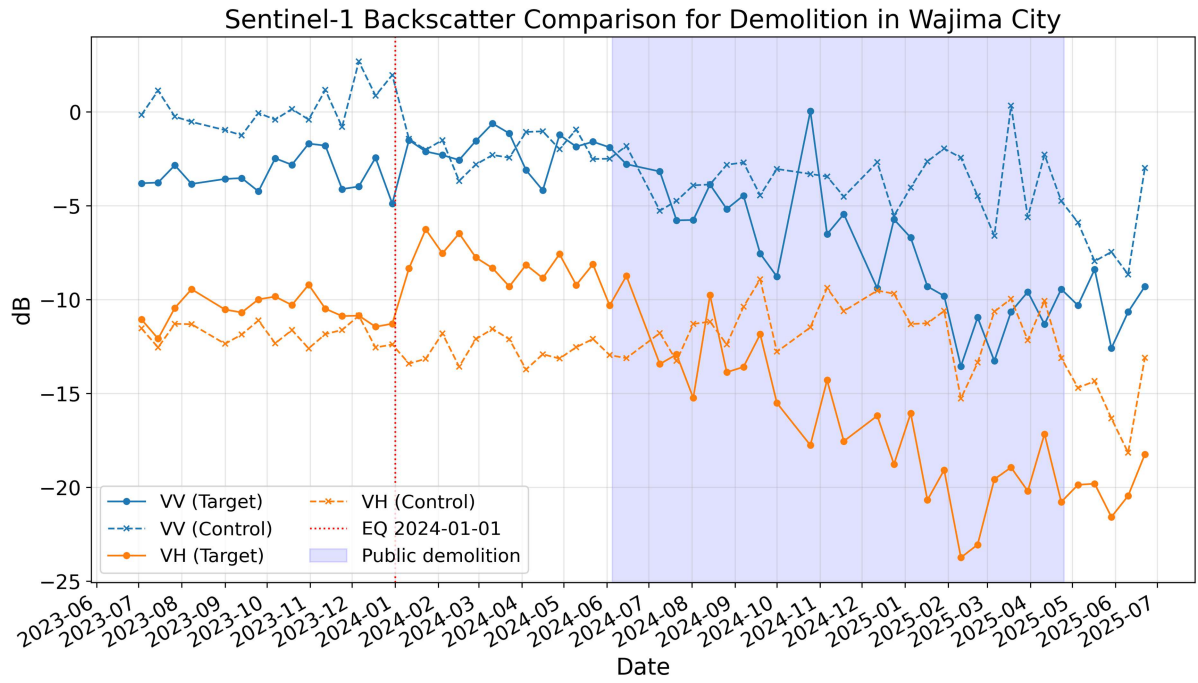


Figure 4. Backscatter coefficients of VV and VH polarizations at the target and control sites in Wajima.

Table 3. Comparison of Sentinel-1 indicators between target and control sites for the demolition site in Wajima City. VV and VH values are in dB. $\bar{x}_T - \bar{x}_C$ denotes the difference between the target and control means. SD_T/SD_C denotes the ratio of their standard deviations.

Indicator	Period	$\bar{x}_T - \bar{x}_C$	SD_T/SD_C
VV	Pre-EQ	-3.53	0.83
VV	Post-EQ	0.04	1.16
VV	Demolition	-3.85	2.25
VV	Post-Demolition	-3.66	0.70
VH	Pre-EQ	1.29	1.32
VH	Post-EQ	4.53	1.58
VH	Demolition	-5.22	2.51
VH	Post-Demolition	-4.67	0.62
Coherence	Pre-EQ	-0.02	2.75
Coherence	Post-EQ	-0.07	1.24
Coherence	Demolition	-0.10	1.81
Coherence	Post-Demolition	-0.00	2.64

site were more moderate: the standard deviation ratio during the demolition period was 2.25 and the difference in mean intersite offset between the pre-earthquake and post-demolition periods was only 0.13 dB (-3.53 vs. -3.66 dB), indicating a relatively small net change.

As shown in Figure 5, the temporal evolution of the coherence at the two sites in Wajima reflects the effects of the earthquake and subsequent processes. The sharp decrease in coherence observed immediately after the earthquake clearly indicates surface disturbances induced by the event. While the control site subsequently recovered to its pre-earthquake level, the target site remained below 0.6 in the following acquisition. The pre-earthquake mean difference in coherence between the two sites was negligible (-0.02), but widened slightly in the post-earthquake period to -0.07 . These persistently low values at the target site can be attributed to fire-damaged buildings.

Coherence at the target site eventually returned to approximately the pre-earthquake level; however, from the onset of publicly funded demolition, values frequently fell below 0.9 and remained consistently lower than those of the control site, with the mean difference expanding to -0.10 . Toward mid-March,

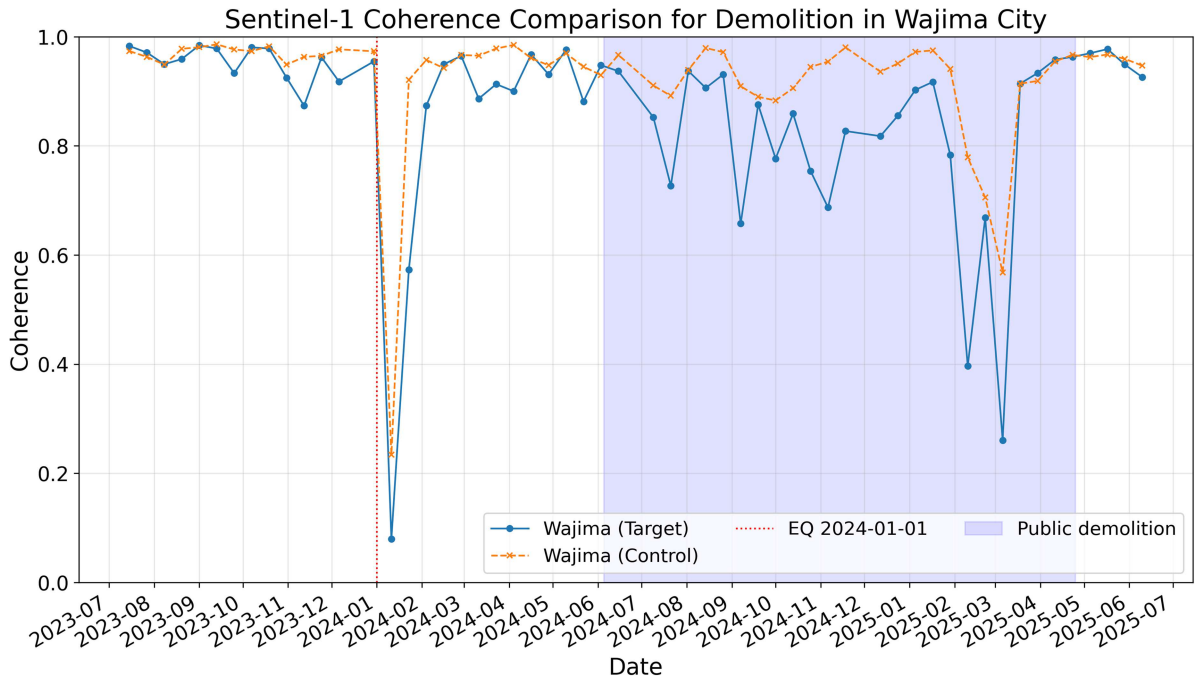


Figure 5. Temporal evolution of coherence at the target and control sites in Wajima. The horizontal axis indicates the later acquisition date of each image pair used in the coherence calculation.

near the completion of the demolition period, the coherence at both sites exceeded 0.9, suggesting stable surface conditions and cessation of demolition activities. A temporary decline in coherence around late February 2025, observed at both the target and control sites, was most likely attributable to the heavy snowfall recorded in February (Weathernews, 2025).

Comparison results for demolition

Sentinel-1 proved more reliable than Sentinel-2 for continuous monitoring of the demolition process. Frequent cloud contamination limited the optical record to approximately 30 usable scenes, whereas SAR data were available at regular 12-day intervals throughout the study period. Of the three optical indices, the NDBI was the most responsive to the earthquake itself, reflecting the post-earthquake reduction in built-up areas; however, it did not exhibit a sustained downward trend during the demolition period. The PL diverged between the target and control sites; however, data gaps in the winter months and possible seasonal effects made interpretation difficult. The NDVI remained nearly constant at the target site; however, this stability meant that the index offered little sensitivity to the progression of demolition.

Among the SAR-based indicators, coherence responded to both the earthquake and the completion of demolition; however, its sensitivity to external disturbances, notably heavy snowfall, undermined its utility for continuous tracking. By contrast, the VH backscatter recorded debris removal as a continuous declining trend, providing the clearest and most interpretable temporal signal for monitoring demolition progress.

Indicator comparison for construction in Suzu city

Sentinel-2 indicators for construction

Figure 6 shows the Sentinel-2 indices for the target and control sites in Suzu. Table 4 summarizes the period-wise comparison of each indicator between the two sites. The analysis period was divided into four phases: pre-earthquake (Pre-EQ), post-earthquake to construction onset (Post-EQ), during construction (Construction), and after construction (Post-Construction); however, because no usable data were available during the Post-EQ period, it was omitted from the table.

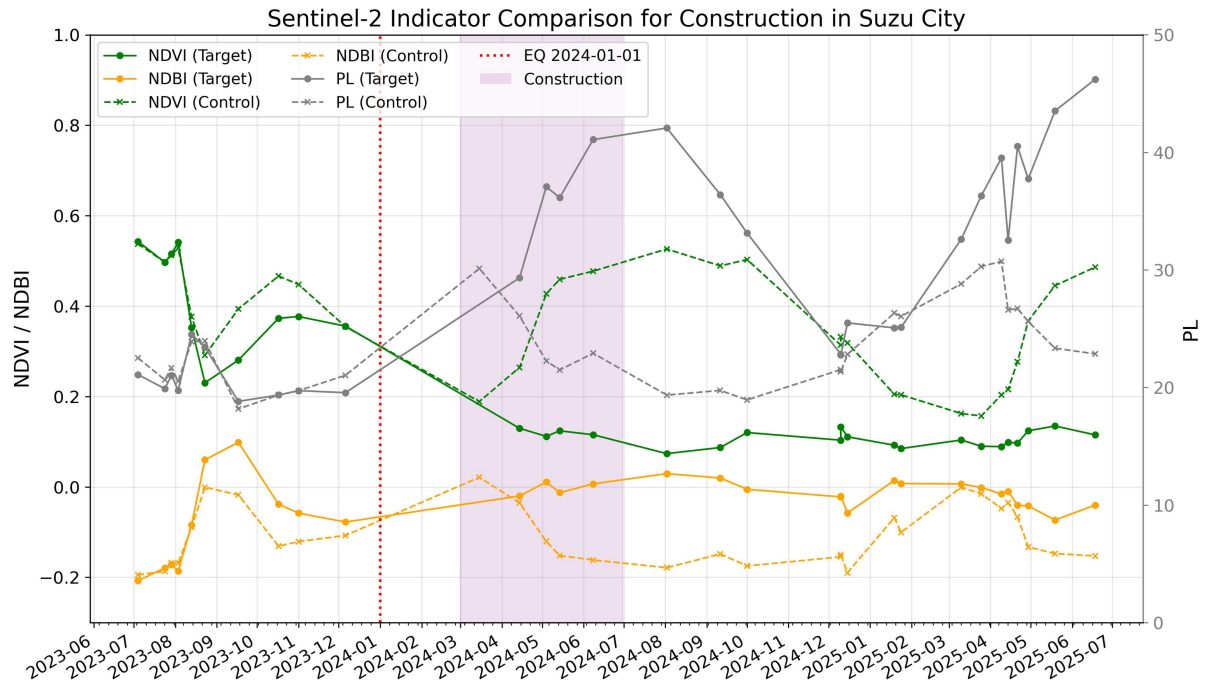


Figure 6. Sentinel-2 indices (NDBI, PL, NDVI) at the target and control sites in Suzu.

Table 4. Comparison of Sentinel-2 indicators between target and control sites for the construction site in Suzu City. $\bar{x}_T - \bar{x}_C$ denotes the difference between the target and control means. SD_T/SD_C denotes the ratio of their standard deviations.

Indicator	Period	$\bar{x}_T - \bar{x}_C$	SD_T/SD_C
NDVI	Pre-EQ	-0.034	1.34
NDVI	Construction	-0.243	0.06
NDVI	Post-Construction	-0.222	0.14
NDBI	Pre-EQ	0.034	1.57
NDBI	Construction	0.086	0.19
NDBI	Post-Construction	0.094	0.47
PL	Pre-EQ	-0.434	0.99
PL	Construction	11.341	1.37
PL	Post-Construction	9.425	2.05

A notable difference emerged in NDVI during the temporary housing construction period. Since the construction site was originally a green area and the control site was selected from a similar environment, both sites exhibited comparable NDVI variations prior to the earthquake, with a mean difference of only -0.034 . As no usable imagery was available immediately after the event, changes in the immediate post-earthquake period could not be captured. However, during the construction phase, a clear divergence appeared: from approximately May 2024, the NDVI at the control site increased above 0.4, whereas that at the target site remained at approximately 0.1. The mean difference expanded to -0.243 during the construction period and persisted at -0.222 after its completion, representing a substantial increase relative to the pre-earthquake baseline. This pattern indicates the loss of the original vegetation and suggests the progression of construction activities.

Similarly, the PL values at the target site increased after April 2024, although the two sites exhibited nearly identical values prior to the earthquake. The mean difference shifted from -0.434 in the pre-earthquake period to 11.341 during construction, confirming a pronounced increase in PL at the target site. This divergence can be interpreted as reflecting the presence of buildings, such as the roofs of

temporary housing, which enhance surface reflectance. However, during the winter months, the PL values decreased, similar to the pattern observed at Wajima, suggesting the influence of seasonal variation.

Finally, the NDBI showed only minor differences between the target and control sites throughout the study period, with the mean difference remaining below 0.1. The smallest difference was observed in the pre-earthquake period (0.034), which increased slightly to 0.086 during construction and 0.094 post-construction. However, the NDBI values at the target site remained near zero, with little variation. These results suggest that NDBI alone is insufficient to confirm the presence of newly developed structures, possibly because of its limited sensitivity to small-scale built-up changes at a 10 m spatial resolution and the material properties of temporary housing, which may not produce the spectral contrast that NDBI is designed to detect. Furthermore, values at the control site exhibited fluctuations potentially attributable to seasonal effects, such as relatively low values (approximately -0.2) in July and values near 0.0 in April. Long-term data analysis may yield further insights; however, the limited number of observations available during the study period remains a limitation.

Sentinel-1 indicators for construction

Figure 7 shows the temporal variations in the VV and VH backscatter coefficients for Suzu, while Table 5 summarizes the comparison of the Sentinel-1 indicators between the two sites for each period.

Before the earthquake, both polarizations exhibited similar fluctuations at the two sites, with mean differences of 2.32 dB for VV and 1.80 dB for VH. After the earthquake but before construction, the mean difference in VV increased to 4.21 dB; however, the standard deviation ratio remained at 0.68, indicating that the target site did not exhibit substantial variability relative to the control site. However, clear differences emerged after the onset of construction. In VV polarization, the mean difference increased markedly to 9.62 dB during the construction period, accompanied by a substantially elevated standard deviation ratio of 3.03, reflecting high variability at the target site. While the control site remained stable at around -13 dB throughout the study period, the target site generally exceeded 0 dB even after construction was completed, with the mean difference further widening to 13.32 dB in the post-construction period. This pattern is consistent with the interpretation that the satellite captured double-bounce scattering from buildings. Although less pronounced than in VV, VH polarization also showed higher values at the target site after the construction period, with the mean difference increasing from 1.58 dB in

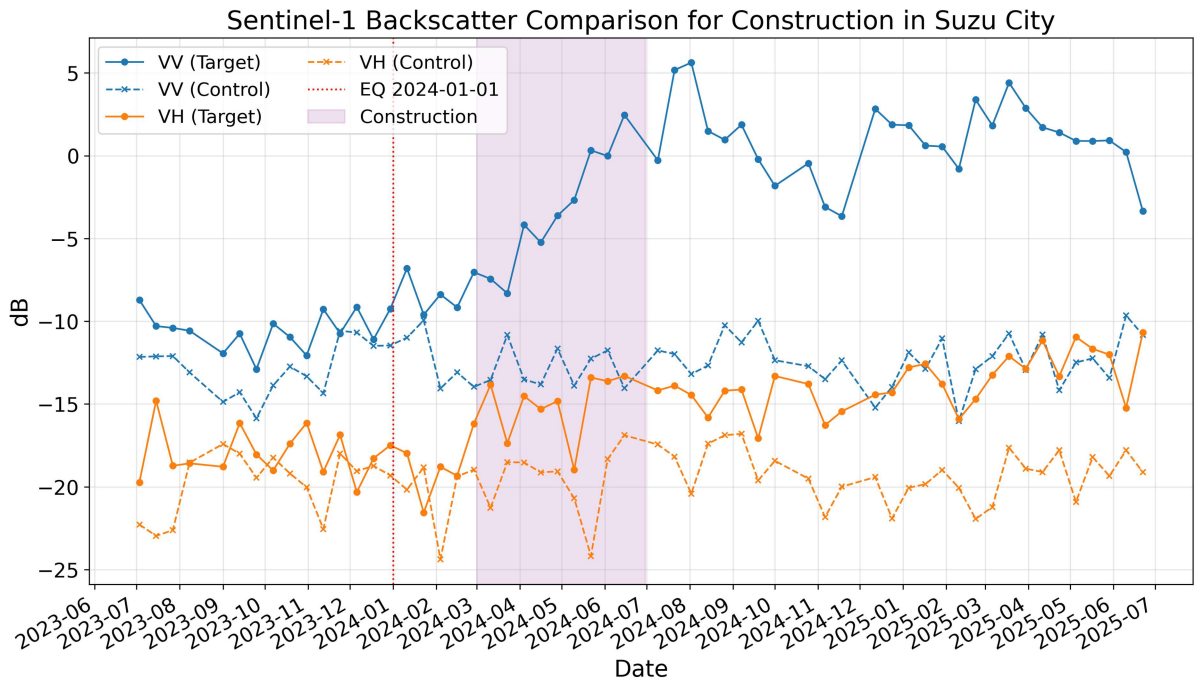


Figure 7. Backscatter coefficients of VV and VH polarizations at the target and control sites in Suzu.

Table 5. Comparison of Sentinel-1 indicators between target and control sites for the construction site in Suzu City. VV and VH values are in dB. $\bar{x}_T - \bar{x}_C$ denotes the difference between the target and control means. SD_T/SD_C denotes the ratio of their standard deviations.

Indicator	Period	$\bar{x}_T - \bar{x}_C$	SD_T/SD_C
VV	Pre-EQ	2.32	0.75
VV	Post-EQ	4.21	0.68
VV	Construction	9.62	3.03
VV	Post-Construction	13.32	1.53
VH	Pre-EQ	1.80	0.79
VH	Post-EQ	1.58	0.85
VH	Construction	4.61	0.91
VH	Post-Construction	5.52	1.10
Coherence	Pre-EQ	-0.09	1.50
Coherence	Post-EQ	0.06	0.92
Coherence	Construction	0.01	1.46
Coherence	Post-Construction	0.02	1.02

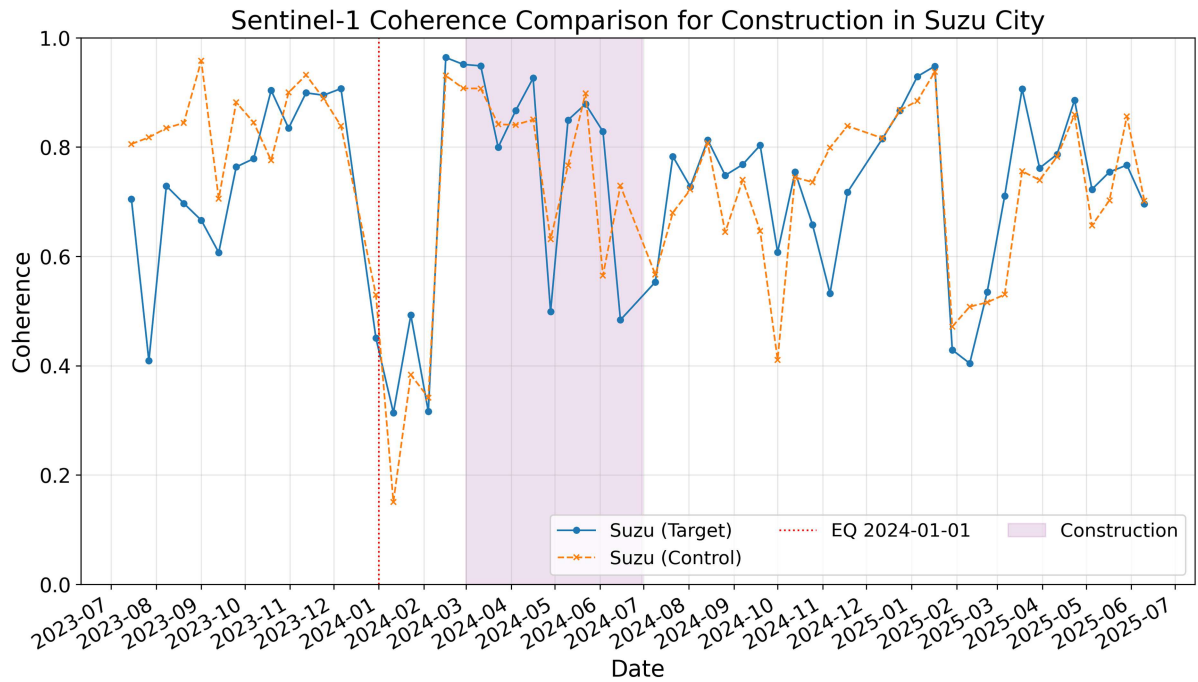


Figure 8. Temporal evolution of coherence at the target and control sites in Suzu. The horizontal axis indicates the later acquisition date of each image pair used in the coherence calculation.

the post-earthquake period to 4.61 dB during construction and 5.52 dB post-construction, suggesting sensitivity to the presence of newly built structures.

Figure 8 shows the temporal coherence variations in Suzu. As observed in Wajima, coherence decreased after earthquakes and heavy snowfall. However, even before the earthquake, coherence values rarely exceeded 0.9, and at the target site, coherence dropped to approximately 0.4 in July 2023, indicating substantial surface variability. As both the target and control sites were located in grassland areas, these fluctuations were likely influenced by vegetation growth. The seasonal pattern of higher coherence in winter and lower values from spring to autumn supports this interpretation. No clear differences between the target and control sites were identified throughout the study period and the mean differences remained close to zero across all phases: -0.09 pre-earthquake, 0.06 post-earthquake, 0.01 during construction, and 0.02 post-construction. This is likely due to the coarse spatial resolution of the coherence analysis (of the order of several tens of meters), which encompasses areas beyond individual buildings.

Comparison results for construction

The limited availability of cloud-free Sentinel-2 imagery constrains the information that optical indices can provide for the construction process. Among the optical indices, the NDVI most clearly captured construction activity, reflecting the loss of original vegetation at the target site, whereas the control site exhibited seasonal greening. The PL increased at the target site after the onset of construction, consistent with the appearance of building roofs, although winter decreases indicate a seasonal component. NDBI showed negligible changes throughout the study period, suggesting that the spectral contrast introduced by the temporary housing materials was insufficient for detection at a 10 m resolution.

In contrast, VV backscatter showed the most pronounced response among the SAR-based indicators, with an increase of approximately 10 dB at the target site, which is attributable to double-bounce scattering from newly constructed buildings. The VH polarization also exhibited higher values after construction, albeit to a lesser extent. In contrast, coherence failed to distinguish the target site from the control site, likely due to the coarse effective resolution of the analysis and the influence of seasonal vegetation changes on the grassland environment.

Selection of Indicators suitable for long-term monitoring and their application to surrounding areas

Building on these results, this section selects the indicators most suitable for long-term post-recovery monitoring, particularly during the early- to mid-recovery phases, and explores their applicability to wider areas to establish satellite-based indicators for assessing building recovery progress.

Sentinel-2, as an optical mission, offers intuitive indicators that contribute to the detection of surface changes. However, optical data are strongly affected by clouds, snow, and seasonal conditions. The limited observation frequency may lead to missed detection of short-term but critical recovery activities, such as demolition or construction. Although this limitation may be mitigated as the Sentinel constellation expands, optical indicators remain less suitable for continuous monitoring. In addition, the sensitivity of certain indices proved insufficient: NDBI did not increase during new building construction in Suzu City, and although NDVI captured vegetation removal, it did not directly indicate the presence of new structures. Rather than serving as the primary monitoring tool, optical data are better suited to subsidiary roles such as confirming land use, distinguishing vegetated areas from residential zones, and corroborating changes detected by SAR.

In contrast, Sentinel-1 SAR enables consistent observations regardless of weather or illumination conditions. The backscatter coefficients, examined by polarization, captured ground changes consistent with debris accumulation, debris removal, and building construction. VH and VV polarizations play complementary roles. VH backscatter, sensitive to volume scattering, effectively captures debris accumulation and removal associated with building demolition, whereas VV backscatter, responsive to double-bounce scattering, detects the emergence of new structures. However, coherence was less suitable for monitoring individual building changes, likely due to its coarse effective spatial resolution.

Based on these findings, the Sentinel-1 backscatter coefficient was considered the most suitable foundational indicator for recovery monitoring. Stable and continuous data availability is invaluable for sustained monitoring, and the use of different polarizations enables physically grounded interpretation, as demonstrated by the complementary roles of VH and VV in capturing the demolition and construction processes, respectively. However, it should be noted that backscatter changes reflect alterations in surface scattering properties rather than the presence of the structures themselves. Therefore, interpretation requires caution. For example, increases in VV backscatter may indicate building construction but could also result from the emergence of other vertical structures, such as trees. In operational applications, ancillary checks, such as verifying that NDVI does not simultaneously increase, are necessary to exclude false detections associated with vegetation growth.

The following subsections examine the applicability of the Sentinel-1 backscatter coefficient by identifying locations with temporal patterns similar to those observed at the selected sites, thereby evaluating its potential for wider area implementation. To extract locations exhibiting similar temporal trends, threshold conditions were defined based on the results obtained in the previous section, thereby enabling the detection of candidate pixels across the surrounding area. However, the threshold values were calibrated

based on the specific conditions observed at the selected target sites, including the magnitude of backscatter changes and the duration of each recovery phase. Application to other disaster contexts or geographic settings would require site-specific calibration, considering the local building characteristics and land cover types.

VH-polarization-based monitoring of the demolition process in Wajima city

This section examines whether pixels exhibiting temporal behaviors similar to those observed at the target site where the public demolition program was conducted could be identified using VH-polarized backscatter. The study area was defined as a 2 km × 2 km square centered on the target site.

As demonstrated in the previous section, the target site exhibited a characteristic three-phase temporal pattern: (1) an increase in VH values after the earthquake, (2) a subsequent gradual decrease, and (3) a sharp decrease corresponding to the onset of public demolition. Based on these phases, threshold conditions were defined for each. Pixels satisfying all the criteria were extracted as candidate locations where damaged buildings were demolished and removed.

Detection patterns of building demolition

Phase 1: increase phase (post-disaster debris generation). Increase requirement: 1–7 Db. To detect an increase in volume scattering caused by structural damage and debris accumulation, the VH backscatter was required to increase by 1–7 dB within the segment. An increase of approximately 5 dB was observed at the target site. The lower bound was set to 1 dB to capture cases in which smaller-scale damage also produced measurable increases in volume scattering, whereas the upper bound of 7 dB was retained to avoid excessive fluctuations unrelated to debris generation.

Phase 2: Gradual Decrease Phase (debris remaining, partial removal). Decrease requirement: ≤ 5 dB. At the target site, a gradual decrease in VH values was observed between the earthquake and the start of the public demolition program. This trend was likely associated with partial debris removal and other small-scale activities prior to full-scale demolition. Based on the magnitude of the changes observed at the target site, a decrease of no more than 5 dB was defined as the threshold for phase 2.

To ensure that the detected pattern corresponded to earthquake-related processes, the onset of phase 2 was required to occur within 60 days after the earthquake. Considering that demolition was expected to commence only after several months, and that it began approximately six months after the event at the target site, a minimum of six samples were required to define this phase. This constraint reflects the assumption that debris accumulation following earthquake damage is followed relatively quickly by settling or partial removal. This prevents the algorithm from identifying unrelated temporal fluctuations that satisfy the amplitude criterion.

Phase 3: sharp decrease phase (public demolition period). Decrease requirement: ≥ 10 dB. During public demolition, a substantial reduction in volume scattering is expected owing to debris removal and dismantling of damaged structures. At the target site, a decrease of approximately 14 dB was observed. Accordingly, a threshold of at least 10 dB was set for phase 3. A minimum of six samples was required to reflect the typical timescale of the demolition process.

To confirm that the detected decrease represents genuine structural removal rather than a temporary fluctuation, the mean VH value during phase 3 was required to fall at least 2.0 dB below the pre-earthquake mean. In densely built residential areas, the complex arrangement of buildings generates substantial volume scattering, and once these structures and the resulting debris are removed, the exposed ground surface produces a lower VH backscatter than the original built-up condition. This condition ensures consistency with the physical expectations.

Detection results of building demolition

Prior to phase detection, the time series data for each pixel were denoised. The mean value of the surrounding 3 × 3 cells was used to suppress local variability, and a Savitzky–Golay filter (window size: five samples; polynomial order: three) was applied (Schafer, 2011) to smooth short-term fluctuations and

After preprocessing, all the time segments that could represent the three-phase pattern were identified. The segment lengths ranged from 20 to 50. All the possible partitions in the three phases were examined in each segment. Only segments satisfying all the threshold conditions were extracted as candidate demolition pixels.

Figure 10 includes a comparison with an aerial photograph acquired by the Geospatial Information Authority of Japan (GSI) on 6 April 2024. The red-outlined areas represent visually confirmed building destruction or exposed soil surfaces. In addition to the fire-affected region, three locations where building collapses were observed were identified. These areas likely represent structural failure owing to the



Figure 9. Spatial distribution of pixels detected by the VH-based demolition detection algorithm in Wajima City, overlaid on an OpenStreetMap basemap. Red shading indicates detected pixels, and the red star marks the target site. Map data © OpenStreetMap contributors, available under the Open Database License (ODbL). The SAR data are georeferenced in WGS 84/UTM zone 53 N (EPSG:32653), and the basemap is rendered in Web Mercator (EPSG:3857) for display.



Figure 10. Comparison between the detected demolition areas and the aerial photograph acquired on 6 April 2024. Areas outlined in red indicate visually confirmed locations where buildings were destroyed and the ground surface was exposed. Areas enclosed in blue denote detections where building collapse could not be confirmed. The aerial photograph, originally acquired by the Geospatial Information Authority of Japan (GSI) in 2024, is shown here with added annotations indicating direction, scale, and detection results.

earthquake, followed by demolition and debris removal, demonstrating that the extraction results aligned with the intended targets.

However, some regions detected by the algorithm did not show clear evidence of building collapse (highlighted in blue in Figure 10). These included residential yards and areas with roofs covered by blue tarps. The residential yard detections represent clear false detections, for which the incorporation of NDVI-based filtering could serve as an effective exclusion strategy. In contrast, the tarp-covered roofs exhibited visible roof damage, suggesting that the building sustained structural damage. Examination of the backscatter time series at these sites revealed a sharp decline around January 2025, as shown in Figure 11, suggesting that buildings were demolished during this period. Further verification using additional datasets or field surveys is necessary to validate this interpretation. Nevertheless, the damage profile at these sites differed substantially from that at the target site, where post-earthquake debris accumulation was the primary indicator; accordingly, these detections also fell outside the intended scope of the algorithm. This suggests that discriminating between different levels of building damage using Sentinel-1 data alone is difficult.

These cases highlight a fundamental limitation of SAR-only approaches: backscatter changes reflect alterations in the surface scattering properties rather than the presence or absence of specific structures. Without complementary information, distinguishing building demolition from other surface modifications remains challenging. Integration with optical data or high-resolution aerial imagery can help reduce such ambiguities and improve detection reliability.

To assess the sensitivity of the detection results to the threshold settings, the minimum decrease required in phase 3 was varied from 8 to 12 dB, while all other parameters were held constant. Table 6 summarizes the results. The number of detected pixels decreased substantially from 498 to 217 in this range, indicating that the extraction was highly sensitive to this parameter. The target site was consistently

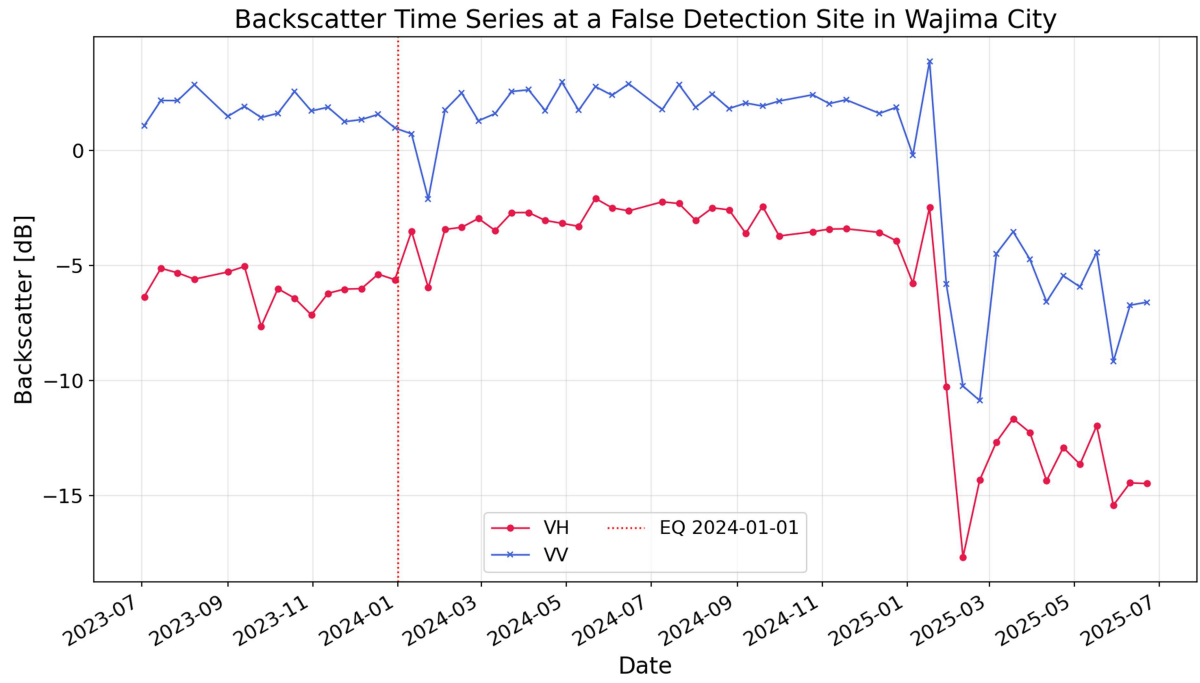


Figure 11. VH and VV backscatter time series at a false detection site in Wajima City.

Table 6. Sensitivity of detection results to the Phase 3 minimum decrease threshold in Wajima City case.

Decrease threshold [dB]	Detected pixels	Target detected
8	498	Yes
10	339	Yes
12	217	Yes

detected under all tested conditions. However, lowering the threshold led to a marked increase in detection at locations beyond the target site, suggesting a corresponding increase in false detections. These results highlight the need for further investigations to establish an appropriate and generalizable threshold-setting procedure.

VV-polarization-based monitoring of the construction process in Suzu city

This section investigates whether pixels exhibiting temporal behaviors similar to those observed at the target site in Suzu, where temporary housing units were constructed after the earthquake, could be identified using VV-polarized backscatter. The study area was defined as a $2 \text{ km} \times 2 \text{ km}$ square centered on the target site. As shown in the previous section, the target site exhibits a characteristic two-phase temporal pattern: (1) a sharp increase in VV values during the construction period and (2) a subsequent long-term stable phase in which high VV values are maintained. Threshold conditions were defined for these phases, and pixels satisfying all criteria were extracted as probable locations for new building construction.

Detection patterns of building construction

Phase 1: increase phase (post-earthquake construction activities). Increase requirement: $\geq 8 \text{ dB}$. To detect changes associated with the construction of temporary housing, periods in which the VV backscatter increased by at least 8 dB relative to the pre-earthquake mean were identified. At the target site, an increase of approximately 10 dB was observed, and construction was completed within approximately four months. To ensure that only earthquake-related construction activities were captured, an increase was required that began within 120 days after the earthquake. The maximum

duration of the increase phase was set to 20 samples (approximately eight months), allowing a sufficient margin beyond the timescale observed at the target site.

Phase 2: Stable Phase (post-construction stability). Mean VV must exceed the pre-earthquake mean by at least 8 dB; standard deviation ≤ 3 dB. After construction, the surface conditions were assumed to remain relatively stable, producing limited fluctuations in the VV backscatter. The following two conditions were imposed during the study period: First, the mean VV value must exceed the pre-earthquake mean by at least 8 dB to confirm that the elevated scattering is attributable to the presence of new structures. Second, the standard deviation must remain at or below 3 dB to ensure that the backscatter level is reasonably stable and not dominated by large fluctuations that may reflect transient surface changes rather than persistent structures. A minimum of ten samples was required for this evaluation. To prevent pixels that exhibit a transient increase during phase 1 but subsequently return to pre-earthquake levels from being falsely detected, phase 2 spans the entire period from the end of the increase phase through the final observation date.

Detection results of building construction

After preprocessing, the pixels were evaluated against the two-phase criteria defined in the previous section, with the earthquake occurrence date (January 1, 2024) as the reference point. Pixels satisfying both phase requirements were classified as locations where construction-related scattering persisted after the earthquakes.

Figure 12 shows the extracted areas overlaid on the base map. The region surrounding the target site, located near the center of the figure, was clearly detected, indicating that the proposed method successfully captured the extent of the constructed buildings. This pattern became more evident when examined alongside an aerial photograph acquired on April 26, 2024, as shown in Figure 13. Four buildings with white roofs were present near the target site, all of which were correctly detected using the algorithm. In addition, the red-highlighted area located on the lower left side contains structures that were also extracted. This location corresponds to another temporary housing construction site (Nikkei Business Publications, 2024), where the buildings had not yet been completed at the time of acquisition. Nevertheless, the method was able to detect the area under construction. The northern red-highlighted area shows a similar pattern; the aerial photograph reveals rows of white-roofed buildings, which are also temporary housing units constructed after the earthquake (Suzu City, 2024). These results demonstrate the effectiveness of the proposed VV-based construction detection approach.

However, the algorithm also detected locations with no evidence of housing construction. This area, marked with a blue frame in Figure 13, is shown in detail in Figure 14, which displays the backscatter time series at this site. A sharp increase in the VV backscatter was observed immediately after the earthquake. Although no newly constructed buildings can be identified in the aerial photograph, some surface changes affecting the VV-polarized scattering appeared to have occurred. Identifying the specific causes requires further investigation using additional data sources or field surveys.

These false detections again underscore the ambiguity inherent in single-sensor SAR analysis, where increases in VV backscatter may result from surface changes unrelated to new constructions. Combining SAR backscatter with other information sources provides additional constraints for distinguishing genuine constructions from other scattering sources.

To examine the threshold sensitivity, the requirement in phase 1 was tested from 6 to 10 dB, as summarized in Table 7. Unlike the Wajima case, the number of detected pixels remained relatively stable across the tested range, decreasing from 249 to 224. This lower sensitivity likely reflects that building construction produces a more pronounced and distinct backscatter increase than the gradual decrease associated with demolition, making the detection less dependent on the precise threshold value. Nevertheless, sensitivity to other parameters was not fully assessed, and further investigation is needed to establish robust and generalizable parameter settings.

Discussion

This study demonstrates that Sentinel-1 backscatter coefficients can effectively capture two key phases of post-disaster building recovery: the demolition of damaged structures and the construction of new

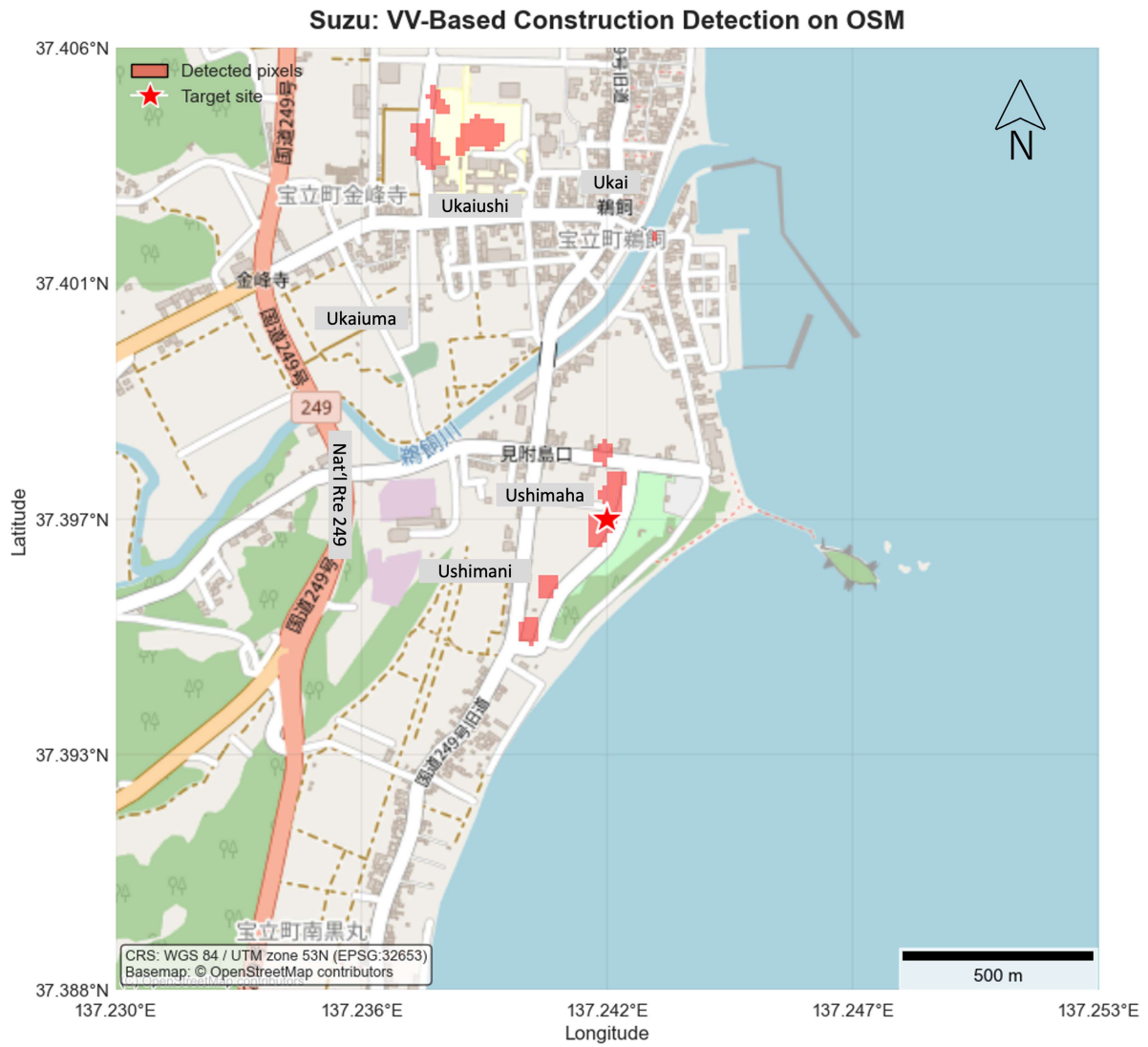


Figure 12. Spatial distribution of pixels detected by the VV-based construction detection algorithm in Suzu City, overlaid on an OpenStreetMap basemap. Red shading indicates detected pixels, and the red star marks the target site. Map data © OpenStreetMap contributors, available under the Open Database License (ODbL). The SAR data are georeferenced in WGS 84/UTM zone 53 N (EPSG:32653), and the basemap is rendered in Web Mercator (EPSG:3857) for display.

buildings. In Wajima City, the VH polarization successfully detected the progressive removal of earthquake-induced debris through a characteristic three-phase temporal pattern. In Suzu City, the VV polarization captured the emergence of temporary housing through a pronounced increase in double-bounce scattering. The complementary roles of these two polarizations provide a comprehensive framework for monitoring physical changes in the built environment during the early to mid-recovery phase. Furthermore, the threshold-based extraction method identified locations exhibiting similar temporal behaviors beyond the target sites, including collapsed buildings in Wajima and additional temporary housing sites in Suzu, which were independently confirmed through aerial photographs.

A key methodological contribution of this study is that freely available Sentinel-1 data, combined with a relatively simple threshold-based detection scheme, can provide systematic and wide-area monitoring of building recovery processes. Unlike optical approaches, which are limited by cloud contamination and seasonal effects, yielding only approximately 30 usable scenes over the study period, SAR-based monitoring offered consistent 12-day acquisitions, regardless of weather or illumination conditions. A comparison between the target and control sites provided quantitative evidence for indicator selection, and subsequent application to the surrounding areas confirmed the scalability of the approach.



Figure 13. Comparison between the detected construction areas and the aerial photograph acquired on 26 April 2024. Areas outlined in red indicate visually confirmed temporary housing units built after the earthquake. Areas enclosed in blue show false detection where no new construction was present. The aerial photograph, originally acquired by the Geospatial Information Authority of Japan (GSI), is shown with added annotations indicating direction, scale, and detection results.

Several limitations should be acknowledged. First, the verification relied on a single-date aerial photograph of each site, and some of the extracted locations did not show clear evidence of demolition or construction. These false detections highlight an inherent limitation of SAR-only approaches; backscatter changes reflect alterations in surface scattering properties rather than the presence or absence of specific structures. In Wajima, false detections included residential yards and buildings covered with blue tarps, where the damage profile differed substantially from the debris accumulation observed at the correctly detected sites. In Suzu, pixels satisfying the two-phase criterion were found at locations where no new construction could be confirmed in the aerial photographs; however, a sharp post-earthquake increase in VV backscatter was observed. These cases demonstrate that discriminating between different types and degrees of surface changes using Sentinel-1 data alone remains challenging.

Second, this study targeted cases involving substantial surface changes, such as debris removal and the construction of new temporary housing units. Partial structural damage, particularly changes below the 10 m spatial resolution of Sentinel-1, may not be detectable. As demonstrated in Wajima, buildings with tarp-covered roofs could not be reliably distinguished from completely collapsed structures, indicating that the method is primarily sensitive to large-scale surface transformations rather than to varying degrees of building damage.

Third, the threshold values and temporal parameters were calibrated based on specific conditions observed at the target sites. The primary aim of this study was to demonstrate the potential of Sentinel-1 backscatter as an indicator for monitoring building recovery, and the parameter settings were tailored accordingly. In addition, the influence of seasonal variations on backscatter, which may arise from changes

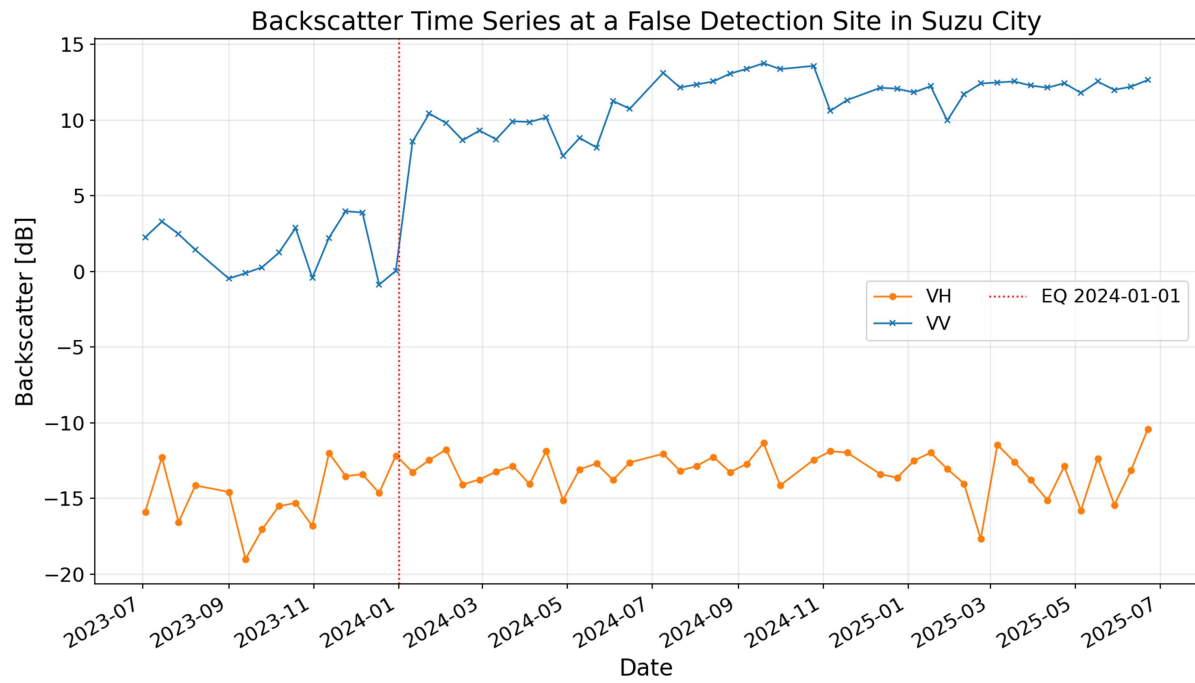


Figure 14. VH and WV backscatter time series at a false detection site in Suzu City.

Table 7. Sensitivity of detection results to the Phase 1 increase threshold in Suzu City case.

Increase threshold [dB]	Detected pixels	Target detected
6	249	Yes
8	247	Yes
10	224	Yes

in vegetation, was not fully considered. Furthermore, the results may be influenced by the specific satellite observation geometry and urban structural characteristics of the study area. In particular, the three-phase VH pattern identified in Wajima was observed in a district characterized by densely aligned wooden row houses with a specific street orientation relative to the satellite look direction. The pre-earthquake backscatter baseline at this site was partly governed by these structural characteristics, and the magnitude of the subsequent VH changes was consequently dependent on this initial condition. In urban areas with different building densities or street configurations, the pre-earthquake baseline would differ, potentially resulting in different magnitudes of VH change even for comparable levels of structural damage. In areas with complex topography, terrain-induced variations in the local incidence angle would further modify the backscatter response. Consequently, the generalizability of these thresholds and patterns to other disaster contexts and geographic settings remains uncertain, and site-specific evaluation would be required before applying the approach established in this study.

Future work should address these limitations in three ways. First, validation should be improved through multi-temporal aerial photographs, ground surveys, or very high-resolution satellite imagery to enable a more rigorous assessment of detection accuracy and reduce reliance on single-date reference data. In addition, applying this method to a larger number of cases and comparing the temporal patterns across different building types and land cover conditions will enhance the robustness of the approach. Second, beyond case-by-case calibration, the adoption of adaptive thresholding methods, such as anomaly detection techniques that automatically identify significant deviations from baseline temporal patterns, should be explored to reduce dependence on manually defined thresholds. Third, the integration of complementary data sources can help reduce false detections and improve interpretability. In particular, NDVI-based filtering could effectively exclude vegetated areas, such as residential yards, that were falsely detected in

Wajima, whereas high-resolution optical imagery could help distinguish tarp-covered buildings from genuinely collapsed structures. Cross-referencing using building footprint databases or construction records can verify whether the detected values correspond to the actual physical activities.

Addressing these challenges and establishing a reliable methodology for time-series monitoring of buildings and infrastructure using the Sentinel series would enable the continuous assessment of reconstruction progress over wide areas. Although changes in buildings alone cannot fully capture the return of people or economic activity, monitoring physical changes in the built environment constitutes an important component for evaluating BBB. The approach presented in this study addresses the demolition and construction dimensions of recovery that represent directly observable indicators among the multiple facets of BBB. Integration with broader BBB assessment frameworks, incorporating socioeconomic data, infrastructure functionality, etc., would enable a more comprehensive evaluation of recovery progress. Furthermore, a systematic analysis of reconstruction processes from past disasters using the proposed methodology can provide valuable insights for planning and improving future recovery strategies.

Conclusion

This study evaluates the potential of Sentinel-series satellites to detect physical changes in buildings during post-disaster recovery, focusing on publicly funded demolition and temporary housing construction following the 2024 Noto Peninsula earthquake. By comparing target sites with control sites of similar pre-earthquake land-use characteristics, we evaluated three indices derived from Sentinel-2 (NDBI, NDVI, and PL), together with the Sentinel-1 backscatter coefficients (VV and VH) and coherence. While optical indicators were affected by cloud contamination and showed limited sensitivity to certain structural changes, Sentinel-1 backscatter provided consistent and physically interpretable signals suitable for long-term monitoring.

Distinct temporal backscatter patterns were identified for each recovery process. In demolition areas, VH polarization captured a three-phase pattern reflecting debris generation, partial removal, and demolition. In construction areas, VV polarization increased rapidly, followed by long-term stability. The complementary roles of VH and VV polarizations provide a comprehensive framework for monitoring the clearance and reconstruction phases of building recovery. Application of threshold-based temporal criteria across wider areas extracted additional collapsed building sites and constructed temporary housing units, thereby demonstrating the scalability of the approach.

Although false detections occurred and threshold calibration was site-specific, the results demonstrated that freely available Sentinel-1 data combined with a threshold-based detection scheme can provide systematic wide-area monitoring of building recovery processes.

The approach of characterizing temporal backscatter patterns associated with demolition and construction can be applied to other post-disaster contexts with similar recovery processes. By offering an operational and scalable means of tracking physical changes in the built environment, this approach can support the continuous assessment of reconstruction progress over wide areas and contribute to the satellite-based evaluation of the BBB, complementing socioeconomic and community-level assessments of recovery.

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HS: Conceptualization; Methodology; Investigation; Formal analysis; Visualization; Writing—original draft. SY: Conceptualization; Writing—review & editing; Supervision. All authors have read and approved the submitted version of the manuscript and agree to be accountable for all aspects of the work.

Author contributions

CRedit: **Hiroaki Sato**: Conceptualization, Formal analysis, Investigation, Methodology, Resources, Visualization, Writing – original draft; **Soichiro Yasukawa**: Conceptualization, Supervision, Writing – review & editing.

Disclosure statement

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Data availability statement

The Sentinel-1 and Sentinel-2 data used in this study are publicly available from the Copernicus Open Access Hub (<https://scihub.copernicus.eu/>) and HyP3 (<https://hyp3-docs.asf.alaska.edu/#web-access>). Processed datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

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